

Zhaoyi Wang

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Education

PhD in Geomatics , Institute of Geodesy and Photogrammetry, ETH Zurich	10/2021 – 12/2025
MSc in Precision Engineering Geodesy , University of Science and Technology of China	9/2018 – 6/2021
BSc in Geomatics , Changsha University of Science and Technology	9/2014 – 6/2018

Professional Experiences

Research Assistant , Geosensors and Engineering Geodesy, ETH Zurich, Switzerland	10/2021 – 12/2025
Research Visiting Student , 3DOM, Fondazione Bruno Kessler (FBK), Trento, Italy	9/2025 – 11/2025
Conducted research on 3D change detection and UAV-based 3D reconstruction	
Compute Vision Algorithm Research Intern , Dilusense Co., Ltd., Hefei, China	3/2021 – 6/2021
Worked on structured light camera calibration and depth estimation	
Summer School Student , International Geoinformatics Summer School, Wuhan University, China	8/2020
Exchange Student , Dept. of Land Surveying and Geo-Informatics, Hong Kong Polytechnic University, China	7/2017

Selected Activities and Achievements

- Best Presentation Award**, ISPRS Geospatial Week – Laser Scanning Workshop, 2023
- National Scholarship for Postgraduates**, University of Science and Technology of China, Top 2%, 2020
- Top Grade Scholarship**, Changsha University of Science and Technology, Top 1% (4×), 2015–2018
- Peer Reviewer** for the *ISPRS J. Photogramm. Remote Sens.*, *Int. J. Appl. Earth Obs. Geoinf.*, *Comput. Urban Sci.*, and *IEEE Lat. Am. Trans.*

Core Skills and Competences

- Research Interests:** LiDAR Point Cloud Processing, Photogrammetric 3D Reconstruction, Multimodal Data and Feature Fusion, 3D Deep Learning, Point Cloud Registration, Landslide Monitoring, and 3D Change Detection
- Programing Languages:** Python, C++, Shell scripting, MATLAB
- Development Tools:** CUDA, Git, LaTeX, Linux
- Frameworks & Libraries:** PyTorch, Open3D, OpenCV, COLMAP
- Professional Software:** Leica Cyclone, RIEGL RiSCAN PRO, FARO SCENE, Agisoft Metashape, CloudCompare, Unity
- Languages:** Mandarin (Native), English (Professional Working Proficiency)
- Extracurricular Activities:** Half-Marathon Running, Skiing, Hiking, Table Tennis

Peer-Reviewed Publications (including in progress)

Z.Y. Wang, P. Trybala, A. Wieser, and F. Remondino. MultiChange3D: A Multi-Scene, Multi-Sensor Dataset for Benchmarking 3D Geometric Change Detection. (Submitted, 2025).

Z.Y. Wang, J.A. Butt, S.Y. Huang, T. Medić, and A. Wieser. Dense 3D Displacement Estimation for Landslide Monitoring via Fusion of TLS Point Clouds and Embedded RGB Images. *Int. J. Appl. Earth Obs. Geoinf.* (Under review, 2025).

Z.Y. Wang, J.A. Butt, S.Y. Huang, N. Meyer, T. Medić, and A. Wieser. An Approach for RGB-Guided Dense 3D Displacement Estimation in TLS-Based Geomonitoring. *ISPRS Ann. (ISPRS Geospatial Week - Laser Scanning Workshop)*, 2025.

Z.Y. Wang, S.Y. Huang, J.A. Butt, Y.Z. Cai, M. Varga, and A. Wieser. Cross-Modal Feature Fusion for Robust Point Cloud Registration with Ambiguous Geometry. *ISPRS J. Photogramm. Remote Sens.*, 2025.

Z.Y. Wang, M. Varga, T. Medić, and A. Wieser. Assessing the Alignment Between Geometry and Colors in TLS Colored Point Clouds. *ISPRS Ann. (ISPRS Geospatial Week - Laser Scanning Workshop)*, 2023.